

121/1
MATHEMATICS
PAPER 1
TIME: 2½ HOURS

SECTION I (50 Marks)

*Answer **ALL** questions in this section in the spaces provided.*

1. Without using mathematical tables or a calculator, evaluate:

$$\frac{-12 \div (-3) \times 4 - (-2)}{-6 \times 2 + (-5)}$$

(3 marks)

2. A cold storage room's temperature was -15°C at 7:00 a.m. The temperature was lowered by 3°C every hour until 11:00 a.m. After a power failure at 11:00 a.m., the temperature rose by 5°C every hour until 2:00 p.m. Find the temperature of the room at 2:00 p.m. (3 marks)

3. Simplify the algebraic expression completely:

$$\frac{3x^2 - 48}{x^2 - 2x - 8}$$

(3 marks)

4. A forex bureau buys and sells foreign currencies at the rates shown below:

Currency	Buying (Ksh)	Selling (Ksh)
1 US Dollar (\$)	128.50	129.20
1 Sterling Pound (£)	162.10	163.40

A businessman arrived in Kenya with £8,500. He converted all his money into Kenya Shillings. While in Kenya, he spent a total of Ksh 650,000. He then converted the remaining balance into US Dollars. Calculate the exact amount of US Dollars he received. (3 marks)

5. Using a ruler and a pair of compasses only, construct a parallelogram $ABCD$ in which $AB = 7$ cm, $AD = 5$ cm, and angle $DAB = 60^{\circ}$. From point D , drop a perpendicular line to meet AB at N . Measure the length of DN . (4 marks)
6. Two similar containers have capacities of 1.5 liters and 12 liters respectively. If the smaller container has a surface area of 450 cm^2 , calculate the surface area of the larger container. (3 marks)
7. The interior angle of a regular polygon is 150° larger than its exterior angle. Determine the number of sides of the polygon. (3 marks)

8. Solve for x in the equation:

$$3^{2x+1} - 10 \cdot 3^x + 3 = 0$$

(3 marks)

9. Given that $\sin(2x + 30^\circ) = \cos(3x)$, where x is an acute angle, find the value of x and hence evaluate $\tan(x + 3^\circ)$. (3 marks)
10. A line L_1 passes through the points $P(-2, 5)$ and $Q(4, -1)$. Another line L_2 is perpendicular to L_1 and passes through the point $(1, 3)$. Find the equation of L_2 in the form $y = mx + c$. (3 marks)
11. The table below shows the masses, to the nearest kg, of 40 students in a class:

Mass (kg)	40 - 44	45 - 49	50 - 54	55 - 59	60 - 64
Frequency	4	8	15	10	3

Calculate the semi-interquartile range of the data.

(4 marks)

12. The position vectors of points A and B are $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j}$ and $\mathbf{b} = -2\mathbf{i} + 8\mathbf{j}$ respectively. A point C divides line AB externally in the ratio $3 : 1$. Find the coordinates of point C . (3 marks)
13. Use the mid-ordinate rule with 4 strips to estimate the area bounded by the curve $y = x^2 + 2$, the x -axis, and the lines $x = 0$ and $x = 4$. (3 marks)
14. Find the range of values of x which satisfy the following simultaneous inequalities and represent the solution on a number line:

$$3 - 2x < 5$$

$$4 - 3x \geq -8$$

(3 marks)

15. A solid cone has a base radius of 7 cm and a slant height of 25 cm. Calculate the total surface area of the cone. (Take $\pi = \frac{22}{7}$). (3 marks)
16. The displacement, s meters, of a particle moving along a straight line after t seconds is given by $s = 2t^3 - 9t^2 + 12t + 4$. Find the acceleration of the particle when it is momentarily at rest. (3 marks)

SECTION II (50 Marks)

Answer any **FIVE** questions from this section.

17. Three merchants, Onyango, Kiprop, and Mwangi, contributed a total of Ksh 1,200,000 to start a wholesale business in Nakuru. Their contributions were in the ratio 2:3:5 respectively.
- (a) Calculate the amount of money contributed by each merchant. (3 marks)
 - (b) At the end of the year, the business realized a profit of Ksh 360,000. They agreed to set aside 25% of the profit for modernizing the store and share the remainder in the ratio of their initial contributions. Calculate the amount of profit received by Mwangi. (3 marks)
 - (c) The following year, Kiprop withdrew his entire profit allocation and reinvested it back into his share capital, while Onyango and Mwangi spent theirs. Calculate Kiprop's new percentage shareholding in the business. (4 marks)
18. A bus left Mombasa for Nairobi, a distance of 500 km, at 7:30 a.m. traveling at an average speed of 80 km/h. At 8:30 a.m., a station wagon left Nairobi for Mombasa traveling along the same road at an average speed of 100 km/h.
- (a) Calculate the distance traveled by the bus before the station wagon started its journey. (2 marks)
 - (b) Determine:
 - (i) The time of day when the two vehicles met. (4 marks)
 - (ii) How far from Nairobi the meeting point was. (2 marks)
 - (c) A modern layout highway patrol car left Nairobi at 9:00 a.m. heading to Mombasa at a constant speed of 120 km/h. Find the distance between the patrol car and the bus when the clock struck 10:30 a.m. (2 marks)
19. The vertices of a triangle ABC are $A(1, 2)$, $B(4, 2)$, and $C(2, 4)$.
- (a) On a Cartesian plane, draw triangle ABC . (1 mark)
 - (b) Triangle $A'B'C'$ is the image of triangle ABC under a reflection in the line $y = -x$. Determine the coordinates of A' , B' , and C' and draw triangle $A'B'C'$ on the same grid. (3 marks)
 - (c) Triangle $A''B''C''$ is the image of triangle $A'B'C'$ under a rotation of $+90^\circ$ about the origin $(0, 0)$. State the coordinates of A'' , B'' , and C'' and plot triangle $A''B''C''$. (3 marks)
 - (d) Describe a single transformation that maps triangle ABC directly onto triangle $A''B''C''$. (3 marks)
20. The figure represents a solid cylinder of radius 10.5 cm from which a smaller coaxial cylinder of radius 3.5 cm and depth 8 cm has been drilled out from the top face. The total height of the original solid cylinder was 20 cm. (Take $\pi = \frac{22}{7}$).
- (a) Calculate the volume of the remaining solid body. (4 marks)

(b) Calculate the total surface area of the remaining solid. (6 marks)

21. Four towns P , Q , R , and S are situated such that town Q is 180 km on a bearing of 060° from town P . Town R is 240 km on a bearing of 150° from town Q . Town S is due South of town P and due West of town R .

(a) Using a scale of 1 cm to represent 40 km, construct a scale drawing showing the relative positions of towns P , Q , R , and S . (4 marks)

(b) From your scale drawing, determine:

(i) The distance between town P and town R . (2 marks)

(ii) The bearing of town R from town P . (1 mark)

(iii) The distance between town P and town S . (2 marks)

(iv) The bearing of town Q from town S . (1 mark)

22. The cross-section of an open drainage trough is modeled mathematically by the curve $y = \frac{1}{2}x^2 - 2x + 3$ for the domain $0 \leq x \leq 4$, where x and y are measured in meters.

(a) Complete the table of values below for the curve: (2 marks)

x	0	0.5	1	1.5	2	2.5	3	3.5	4
y	3	2.13		1.13	1		1.5	2.13	3

(b) Use the trapezoidal rule with 8 strips to estimate the cross-sectional area bounded by the curve, the x -axis, and the vertical lines $x = 0$ and $x = 4$. (3 marks)

(c) Use integration to find the exact area of the cross-section. (3 marks)

(d) Calculate the percentage error introduced by using the trapezoidal rule approximation. (2 marks)

23. Two vertical observation towers, A and B , stand on a level horizontal clearing. Tower A is 45 meters high. From the top of tower A , the angle of depression of the base of tower B is 30° , and the angle of elevation of the top of tower B is 20° .

(a) Calculate the horizontal distance between the two towers, giving your answer to 2 decimal places. (3 marks)

(b) Calculate the total height of tower B . (3 marks)

(c) From a window located 15 meters up from the base of tower A , a ranger looks down at an injured wild animal resting at the base of tower B . Calculate the angle of depression of the animal from this window. (4 marks)

24. The boundaries of a piece of community farming land were surveyed, and the measurements entered in a surveyor's field book as shown below. (The base line is $AG = 280$ meters, and all offsets are measured in meters).

To G (280)		
To F (60)	220	80 to E
	160	
To H (50)	100	40 to C
	40	
From A		

- (a) Using a scale of 1 cm to represent 20 meters, draw an accurate layout map of the farming land. (3 marks)
- (b) Calculate the total area of the farming land in hectares. (7 marks)

END OF EXAMINATION!!!!!!